```
# Loading all the tools I need to continue first
import requests
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
from scipy import stats
import numpy as np

print("Libraries loaded successfully.")
```

```
Libraries loaded successfully.
```

```
# Scryfall updates their download link daily so I can't hardcode it.
# This first request asks Scryfall where today's file is located.
bulk_info_url = "https://api.scryfall.com/bulk-data"
response = requests.get(bulk_info_url)
bulk_data = response.json()

# From the response, I'm grabbing just the "default_cards" file.
# This is the one that has one entry per card printing with price data.
default_cards_entry = next(
    item for item in bulk_data["data"] if item["type"] == "default_cards"
)

# Pulling out just the actual download URL from that entry.
download_url = default_cards_entry["download_uri"]
print(f"Download URL found: {download_url}")

# Now downloading the full card dataset — this can take about 30-60 seconds.
# Roughly 200-300MB of data.
print("Downloading card data...")
cards_response = requests.get(download_url)

# Converting the raw downloaded data into a Python list I can work with.
# Each item in the list is one card with all its details.
cards_data = cards_response.json()

# Counts how many items are in the list or how many cards we got.
print(f"Total cards downloaded: {len(cards_data):,}")
```

```
Download URL found: https://data.scryfall.io/default-cards/default-cards-20260402090808.json
Downloading card data...
Total cards downloaded: 113,185
```

```
# Loading the card list into a DataFrame — rows = cards, columns = attributes
df_raw = pd.DataFrame(cards_data)

# Verifying what we have
print(f"Shape: {df_raw.shape}")  # rows x columns
print(f"\nSample cmc values:\n{df_raw['cmc'].head(10)}")
print(f"\nSample prices values:\n{df_raw['prices'].head(3)}")
```

```
Shape: (113185, 88)

Sample cmc values:
0    0.0
1    6.0
2    2.0
3    0.0
4    3.0
5    1.0
6    0.0
7    2.0
8    5.0
9    1.0
Name: cmc, dtype: float64

Sample prices values:
0    {'usd': '0.36', 'usd_foil': '0.45', 'usd_etche...
1    {'usd': '0.56', 'usd_foil': '2.92', 'usd_etche...
2    {'usd': '0.18', 'usd_foil': '1.77', 'usd_etche...
Name: prices, dtype: object
```

```
# Pulls the 'usd' price out of the nested prices dictionary into its own column
df_raw["price_usd"] = df_raw["prices"].apply(
```

```
    lambda x: float(x["usd"]) if x.get("usd") is not None else None
)

# Keep only the columns we actually need
df = df_raw[["name", "cmc", "price_usd", "rarity", "lang"]].copy()

print(f"Before filtering: {len(df):,} rows")

# English cards only
df = df[df["lang"] == "en"]
print(f"After English filter: {len(df):,} rows")

# Drops rows missing cmc or price
df = df.dropna(subset=["cmc", "price_usd"])
print(f"After dropping nulls: {len(df):,} rows")

# Remove cards with $0 price
df = df[df["price_usd"] > 0]
print(f"After removing $0 prices: {len(df):,} rows")

# Cap cmc at 17
df = df[df["cmc"] <= 17]
print(f"After capping cmc at 17: {len(df):,} rows")

# Show stats
print(f"\nBasic statistics:\n{df[['cmc', 'price_usd']].describe()}")
```

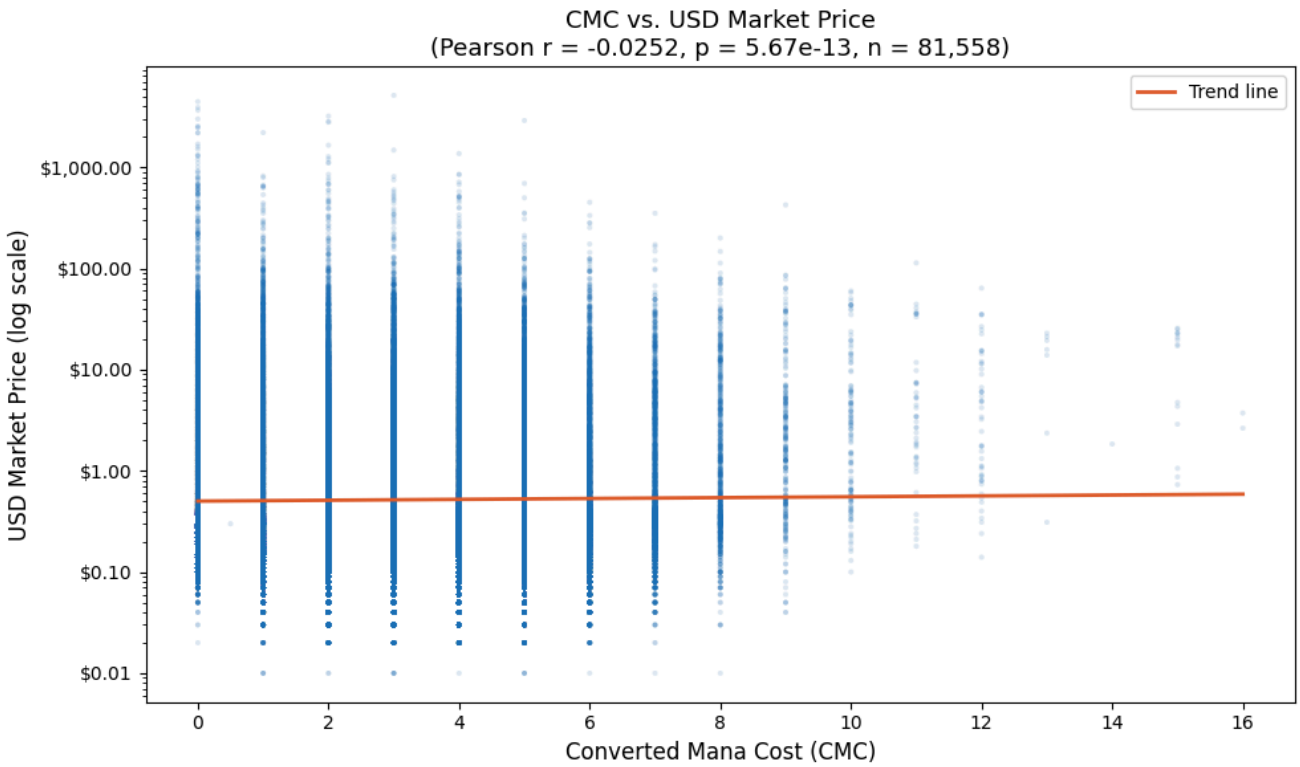
```
Before filtering: 113,185 rows
After English filter: 110,676 rows
After dropping nulls: 81,559 rows
After removing $0 prices: 81,559 rows
After capping cmc at 17: 81,558 rows

Basic statistics:
                cmc     price_usd
count  81558.000000  81558.000000
mean       2.919389      4.686892
std        1.966274     53.055738
min        0.000000      0.010000
25%        2.000000      0.150000
50%        3.000000      0.310000
75%        4.000000      1.280000
max       16.000000   5142.020000
```

```
r, p_value = stats.pearsonr(df["cmc"], df["price_usd"])

alpha = 0.05

print("=" * 50)
print("PEARSON CORRELATION RESULTS")
print("=" * 50)
print(f"Pearson r:    {r:.4f}")
print(f"P-value:      {p_value:.4e}")
print(f"Alpha:        {alpha}")
print(f"Sample size:  {len(df):,}")
print()

if p_value < alpha:
    print("CONCLUSION: Reject the null hypothesis.")
    print("There IS a statistically significant correlation")
    print("between cmc and USD market price.")
else:
    print("CONCLUSION: Fail to reject the null hypothesis.")
    print("There is NO statistically significant correlation")
    print("between cmc and USD market price.")
```

```
==================================================
PEARSON CORRELATION RESULTS
==================================================
Pearson r:    -0.0252
P-value:      5.6698e-13
Alpha:        0.05
Sample size:  81,558

CONCLUSION: Reject the null hypothesis.
There IS a statistically significant correlation
between cmc and USD market price.
```

```
fig, ax = plt.subplots(figsize=(10, 6))
```

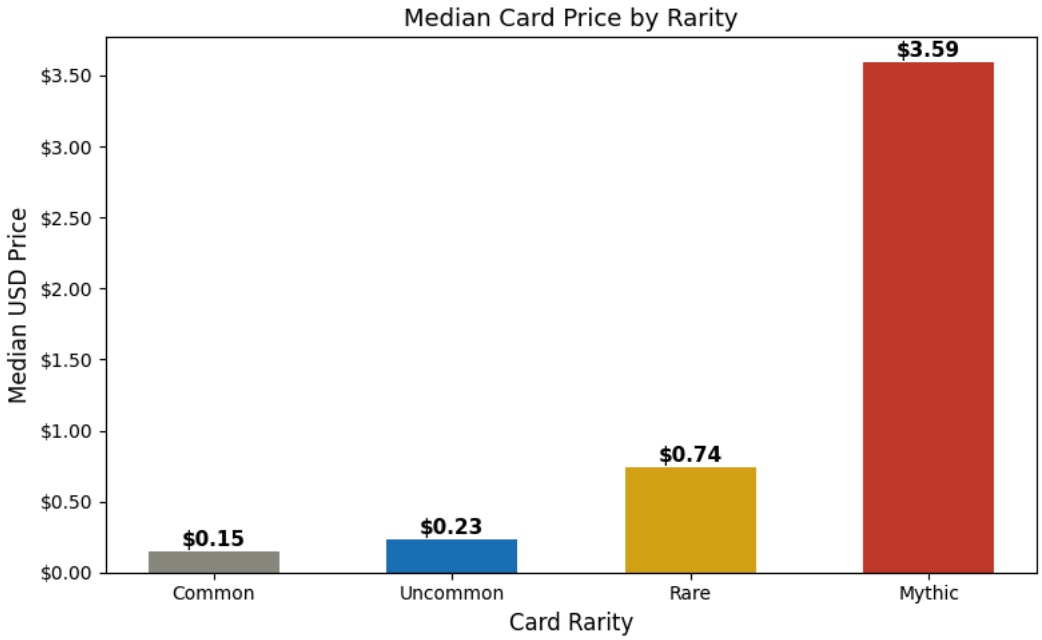
```
# Plotting each card as a dot
# Overlapping dots so they don't just look like a solid blob
ax.scatter(
    df["cmc"],
    df["price_usd"],
    alpha=0.15,
    s=8,
    color="#1a6eb5",
    edgecolors="none"
)

# Fit and draw a trend line
m, b = np.polyfit(df["cmc"], np.log10(df["price_usd"] + 0.01), 1)
x_line = np.linspace(df["cmc"].min(), df["cmc"].max(), 200)
y_line = 10 ** (m * x_line + b)
ax.plot(x_line, y_line, color="#e05a2b", linewidth=2, label="Trend line")

# Log scale because prices span from $0.01 to $5,000+
ax.set_yscale("log")
ax.yaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f"${x:,.2f}"))
ax.set_xlabel("Converted Mana Cost (CMC)", fontsize=12)
ax.set_ylabel("USD Market Price (log scale)", fontsize=12)
ax.set_title(
    f"CMC vs. USD Market Price\n(Pearson r = {r:.4f}, p = {p_value:.2e}, n = {len(df):,})",
    fontsize=13
)
ax.legend()
plt.tight_layout()
plt.savefig("scatter_cmc_vs_price.png", dpi=150)
plt.show()
print("Scatter plot saved.")
```




```
Scatter plot saved.
```

```
rarity_order = ["common", "uncommon", "rare", "mythic"]
rarity_labels = ["Common", "Uncommon", "Rare", "Mythic"]
rarity_colors = ["#888780", "#1a6eb5", "#d4a017", "#c0392b"]

# Calculate median price and card count for each rarity
avg_price_by_rarity = (
    df[df["rarity"].isin(rarity_order)]
    .groupby("rarity")["price_usd"]
    .median()
    .reindex(rarity_order)
```

```
)

count_by_rarity = (
    df[df["rarity"].isin(rarity_order)]
    .groupby("rarity")["price_usd"]
    .count()
    .reindex(rarity_order)
)

fig, ax = plt.subplots(figsize=(8, 5))
bars = ax.bar(
    rarity_labels,
    avg_price_by_rarity.values,
    color=rarity_colors,
    edgecolor="none",
    width=0.55
)

# Add the dollar value label on top of each bar
for bar, val in zip(bars, avg_price_by_rarity.values):
    ax.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + 0.01,
        f"${val:.2f}",
        ha="center",
        va="bottom",
        fontsize=11,
        fontweight="bold"
    )

ax.set_xlabel("Card Rarity", fontsize=12)
ax.set_ylabel("Median USD Price", fontsize=12)
ax.set_title("Median Card Price by Rarity", fontsize=13)
ax.yaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f"${x:.2f}"))
plt.tight_layout()
plt.savefig("bar_price_by_rarity.png", dpi=150)
plt.show()
print("Bar chart saved.")
```




```
Bar chart saved.
```

```
# Results from different calculations and visuals
print("=" * 55)
print("Summary of Statistical Analysis")
print("=" * 55)
print(f"Dataset size (after cleaning):  {len(df):,} cards")
print(f"Pearson r:                       {r:.4f}")
print(f"P-value:                         {p_value:.4e}")
print(f"Alpha:                           {alpha}")
print(f"Statistically significant:       {'Yes' if p_value < alpha else 'No'}")
print()
```

```
print("Median price by rarity:")
for rarity, label in zip(rarity_order, rarity_labels):
    if rarity in avg_price_by_rarity.index:
        n = count_by_rarity[rarity]
        med = avg_price_by_rarity[rarity]
        print(f"  {label:<12} ${med:.2f}  (n={n:,})")
print()

if abs(r) < 0.1:
    strength = "negligible"
elif abs(r) < 0.3:
    strength = "weak"
elif abs(r) < 0.5:
    strength = "moderate"
else:
    strength = "strong"

direction = "positive" if r > 0 else "negative"
print(f"Correlation strength: {strength} and {direction} (r = {r:.4f})")
print(f"CMC explains approximately {r**2 * 100:.2f}% of variance in price.")
```

```
======================================================
Summary of Statistical Analysis
======================================================
Dataset size (after cleaning):  81,558 cards
Pearson r:                       -0.0252
P-value:                         5.6698e-13
Alpha:                           0.05
Statistically significant:       Yes

Median price by rarity:
  Common       $0.15  (n=26,728)
  Uncommon     $0.23  (n=21,202)
  Rare         $0.74  (n=27,603)
  Mythic       $3.59  (n=5,776)

Correlation strength: negligible and negative (r = -0.0252)
CMC explains approximately 0.06% of variance in price.
```

Start coding or generate with AI.